

The least non-divisor of the middle binomial coefficient

On Erdős problem #731: a lower bound of shape $\exp(c\sqrt{\log n})$,
and the program for the matching upper bound

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Abstract

Let $L(n)$ denote the least positive integer not dividing $\binom{2n}{n}$. Erdős asked (#731 in the Erdős problem catalogue, from Erdős–Graham–Ruzsa–Straus 1975) for a “reasonable” $f(n)$ with $L(n) \sim f(n)$ for almost all n . We prove unconditionally that for every $\varepsilon > 0$, for almost all n ,

$$L(n) > \exp((c - \varepsilon)\sqrt{\log n}), \quad c = \sqrt{\log 2} = 0.8325\dots,$$

with a power-of- $\exp(\sqrt{\log})$ saving in the exceptional count, and the constant is sharp for the method: the first-moment threshold for the dominant mechanism sits at the same c , so the union bound meets the first-moment threshold. In particular $L(n)$ is *sub-polynomial but super-polylogarithmic* almost always: proved from below at the sharp constant, and conjecturally $\exp((\sqrt{\log 2} + o(1))\sqrt{\log n})$ — a function of the shape $\exp(\Theta(\sqrt{\log n}))$, not a power of n . Deep-scan numerics (exhaustive to 10^8 , sampled to 10^{35}) verify the law with the sharp constant: after removing the explicit second-order term $\frac{1}{2} \log \sqrt{a}$ ($\approx \frac{1}{4} \log \log n$) the measured constant descends 0.935, 0.894, 0.859, 0.843 toward $\sqrt{\log 2} = 0.8326$, and a parameter-free independent-rail model predicts the median and the 5th/95th percentiles prime-for-prime in 14 of 15 decades. The matching upper bound reduces to a *pairwise* decorrelation statement about balanced primes and is conjectured with the same constant. The dominant mechanism is not the prime-power channel one might expect: for 81–89% of n the minimum is achieved by a *balanced prime* — a prime p with every base- p digit of n below $p/2$, so that $p \nmid \binom{2n}{n}$ by Kummer’s theorem and $L(n) \leq p$.

1 Setup: the two channels

Throughout, p is prime, and for $n \geq 1$ we write the base- p expansion $n = \sum_{i < D} d_i p^i$ with $D = D_p(n) = \lfloor \log n / \log p \rfloor + 1$ digits. By Kummer’s theorem [2], $v_p(\binom{2n}{n})$ equals the number of carries in the base- p addition $n + n$. Two elementary consequences frame everything:

- (i) (*Balanced primes.*) Call n *p-balanced* if every base- p digit satisfies $d_i \leq \frac{p-1}{2}$. Doubling a balanced expansion produces no carries, and conversely a digit $\geq \frac{p+1}{2}$ forces a carry; hence $p \nmid \binom{2n}{n}$ iff n is *p-balanced* (odd p). If n is *p-balanced* then $L(n) \leq p$.
- (ii) (*Digit sandwich.*) Writing $B^+(n, p) = \#\{i : d_i \geq \frac{p+1}{2}\}$ and $B^-(n, p) = \#\{i : d_i \geq \frac{p-1}{2}\}$, every carry requires an incoming digit $\geq \frac{p-1}{2}$ and every digit $\geq \frac{p+1}{2}$ creates a carry, so

$$B^+(n, p) \leq v_p\left(\binom{2n}{n}\right) \leq B^-(n, p).$$

The least non-divisor of any integer is a prime power, so $L(n) = \min_p p^{v_p(\binom{2n}{n})+1}$, and $L(n) \leq T$ forces one of:

- (I) some prime $p \leq T$ has $v_p = 0$: n is p -balanced (*balanced channel*);
- (II) some prime power $p^a \leq T$ with $a \geq 2$ has $v_p \leq a - 1$, hence by (ii) at most $a - 1$ digits of n base p lie in the top range $\{\frac{p+1}{2}, \dots, p-1\}$ (*valuation channel*).

2 The lower bound

Lemma 1 (balanced count, leading-digit form). *For odd p and $N \geq p$,*

$$\#\{1 \leq n \leq N : n \text{ } p\text{-balanced}\} \leq 4 N^{1-\beta_p}, \quad \beta_p := \frac{\log \frac{2p}{p+1}}{\log p}.$$

Proof. Let $k = \lfloor \log N / \log p \rfloor$ and let $D \geq 1$ be the leading base- p digit of N , so $N \geq D p^k$. A balanced $n \leq N$ has at most $\min(D+1, \frac{p+1}{2})$ choices for its digit at position k and at most $\frac{p+1}{2}$ for each of the k lower positions. If $D+1 \leq \frac{p+1}{2}$ the count is at most $(D+1) \left(\frac{p+1}{2}\right)^k \leq 2D p^k \left(\frac{p+1}{2p}\right)^k \leq 2N \left(\frac{p+1}{2p}\right)^k$; otherwise the count is at most $\left(\frac{p+1}{2}\right)^{k+1} \leq 2D \left(\frac{p+1}{2}\right)^k \leq 2N \left(\frac{p+1}{2p}\right)^k$ again. Finally $\left(\frac{p+1}{2p}\right)^k \leq \frac{2p}{p+1} N^{-\beta_p} \leq 2N^{-\beta_p}$ since $k \geq \log N / \log p - 1$. The point of the leading-digit truncation is that the loss is an absolute constant — *not* a factor of p — and this is what makes the union bound sharp. $\square$

Lemma 2 (few big digits). *For odd p , $N \geq p$, and $m \geq 0$,*

$$\#\{1 \leq n \leq N : B^+(n, p) \leq m\} \leq \frac{p+1}{2} N^{1-\beta_p} (D+1)^m, \quad D = \lceil \log N / \log p \rceil.$$

Proof. A string with at most m digits in the top range (of size $\frac{p-1}{2}$) and the rest in the bottom range (of size $\frac{p+1}{2}$) is counted by $\sum_{j \leq m} \binom{D}{j} \left(\frac{p-1}{2}\right)^j \left(\frac{p+1}{2}\right)^{D-j} \leq \left(\frac{p+1}{2}\right)^D \sum_{j \leq m} \binom{D}{j} \leq \left(\frac{p+1}{2}\right)^D (D+1)^m$, and $\left(\frac{p+1}{2}\right)^D \leq \frac{p+1}{2} N^{1-\beta_p}$ as in Lemma 1. (This is the counting genre of the proof of Lemma 3.1 in [3].) $\square$

Theorem 3 (lower bound, sharp constant). *Let $c = \sqrt{\log 2}$. For every $\varepsilon \in (0, c/2)$ there is $\delta = \delta(\varepsilon) > 0$ such that*

$$\#\left\{n \leq N : L(n) \leq \exp((c - \varepsilon)\sqrt{\log N})\right\} \ll N \exp(-\delta\sqrt{\log N}).$$

In particular $L(n) > \exp((c - \varepsilon)\sqrt{\log n})$ for almost all n .

Proof. Set $a = \log 2 \cdot \log N$ (so $c^2 \log N = a$), $t = (c - \varepsilon)\sqrt{\log N}$, and $T = e^t$.

Channel (I). By Lemma 1 and $\beta_p \log p = \log 2 - \log(1 + \frac{1}{p}) \geq \log 2 - \frac{1}{p}$, the exceptional count is at most

$$\sum_{3 \leq p \leq T} 4 N^{1-\beta_p} \leq 4 \pi(T) \cdot N \cdot \max_{3 \leq p \leq T} \exp(-\beta_p \log N) \leq 4 T N \exp\left(-\frac{(\log 2 - \frac{1}{T}) \log N}{t}\right),$$

using $\beta_p \log N \geq (\log 2 - \frac{1}{p}) \log N / \log p \geq (\log 2 - \frac{1}{T}) \log N / t$ for all $p \leq T$ (the map $p \mapsto (\log 2 - \frac{1}{p}) / \log p$ increases only on $[3, 3.2)$ and decreases thereafter, so its minimum on $[3, T]$ is attained at $p = T$ once $T \geq 9$). The exponent is

$$t - \frac{a}{t} + \frac{\log N}{Tt} = \frac{t^2 - a}{t} + o(1) = -\frac{(2c\varepsilon - \varepsilon^2) \log N}{(c - \varepsilon)\sqrt{\log N}} + o(1) \leq -3\delta\sqrt{\log N}$$

for $\delta = \delta(\varepsilon) > 0$ and N large. Note the single power of T from $\pi(T)$: Lemma 1's constant loss (rather than a factor p) is exactly what permits the sharp threshold $t^2 = a$, i.e. $c = \sqrt{\log 2}$.

Channel (II). A pair (p, a) with $a \geq 2$, $p^a \leq T$ forces $p \leq T^{1/2}$, i.e. $u := \log p \leq t/2$, and $m := a - 1 \leq t/u$. By Lemma 2 the count for one pair is at most $\frac{p+1}{2} N^{1-\beta_p} (D+1)^m$ with $D \leq 1 + \log N / \log 3$, so, per pair, the exponent relative to N is at most

$$u + \frac{2t \log \log N}{u} - \left(\log 2 - \frac{1}{p} \right) \frac{\log N}{u} = u + \frac{2t \log \log N}{u} - \frac{a}{u} + \frac{\log N}{pu}.$$

For $p \geq 3$ one has $\frac{\log N}{pu} \leq \frac{1}{3} \cdot \frac{\log N}{u} \leq \frac{a}{2u}$, using simply $\frac{1}{p} \leq \frac{1}{3} \leq \frac{\log 2}{2}$, so the saving retains at least $\frac{a}{2u}$. Since $u \leq t/2$,

$$\frac{a}{2u} \geq \frac{a}{t} = \frac{c^2}{c - \varepsilon} \sqrt{\log N} \geq (c + \varepsilon) \sqrt{\log N},$$

while the positive terms total at most $\frac{t}{2} + \frac{2t \log \log N}{u} \leq \frac{c-\varepsilon}{2} \sqrt{\log N} + O(\sqrt{\log N} \frac{\log \log N}{u})$. For $u \geq (\log \log N)^2$ the second term is $o(\sqrt{\log N})$ and the margin is $\geq (c + \varepsilon - \frac{c-\varepsilon}{2}) \sqrt{\log N} (1 - o(1)) \geq \frac{\varepsilon}{2} \sqrt{\log N}$; for $u < (\log \log N)^2$ the saving $\frac{a}{2u} \geq \frac{a}{2(\log \log N)^2}$ dwarfs every other term outright. Hence each pair contributes at most $N \exp(-\frac{\varepsilon}{2}(1 - o(1))\sqrt{\log N})$, and here the pair count must *not* be relaxed to T : the true count is $\pi(T^{1/2}) \cdot \frac{t}{\log 3} \leq e^{t/2} t$, and since $t/2 = \frac{c-\varepsilon}{2} \sqrt{\log N}$ the total is at most $N t \exp(\frac{t}{2} - \frac{\varepsilon}{2}(1 - o(1))\sqrt{\log N}) \leq N \exp(-(\frac{\varepsilon}{2} - o(1))\sqrt{\log N})$, which closes with $\delta = \varepsilon/6$. (The relaxation to T was harmless at weaker constants and fails exactly at the sharp one, where channel (II)'s margin is thinner than $\log T$.)

Finally, pairs with $p = 2$: by Kummer $v_2(\binom{2n}{n}) = s_2(n)$, the binary digit sum, so $v_2 \leq a - 1 \leq t / \log 2$ forces $s_2(n) = O(\sqrt{\log N})$, and $\#\{n \leq N : s_2(n) \leq m\} \leq \sum_{j \leq m} \binom{D}{j} \leq (D+1)^m = \exp(O(\sqrt{\log N} \log \log N)) = N^{o(1)}$, absorbed.

Summing the two channels gives the stated bound; replacing $\log N$ by $\log n$ costs only the dyadic decomposition $n \in (N/2, N]$. $\square$

Remark 4. *The constant is sharp: the expected number of balanced primes $p \leq e^t$ for a random $n \leq N$ crosses 1 at $t = \sqrt{a}$, the same threshold, so the union bound meets the first moment and no further constant improvement is possible in Theorem 3. A crude version of Lemma 1 with the ceiling $\lceil \log N / \log p \rceil$ loses a factor as large as p and caps the method at $\sqrt{\log 2/2}$; the leading-digit truncation is the entire difference.*

Remark 5 (unconditional facts for every n , and the fluctuation picture). (i) $L(n) \geq 3$ always, and $L(n) = 3$ exactly when $2n$ has only digits 0, 2 in base 3; such n number $2^k - 1$ below 3^k , so $\liminf L(n) = 3$ on a set of counting dimension $\log 2 / \log 3$. (ii) For every large n , every prime in $(\frac{2n}{3}, n]$ is a non-divisor, so $L(n) \leq n^{1+o(1)}$ trivially; using primes in the intervals $(\frac{2n}{2k+1}, \frac{n}{k}]$ together with primes in short intervals [5] gives $L(n) \leq n^{1/(2-\theta)+o(1)} = n^{0.678\dots}$ for

all large n , with $\theta = 0.525$ the short-interval exponent. (iii) Around the almost-all law the fluctuations do not vanish: the independent-rail model gives a minimum-type (Gumbel) limit for $\log L(n) - \sqrt{a} - \frac{1}{2} \log \sqrt{a}$ with bounded scale — $L(n)$ is pinned within a bounded multiplicative window of the law on that model, with the 5th–95th percentile span a constant factor ≈ 9 across fifteen measured decades.

3 The matching upper bound: program and conjecture

Conjecture 6. For every $\varepsilon > 0$, almost all n possess a balanced prime $p \leq \exp((c + \varepsilon)\sqrt{\log n})$; consequently $L(n) = \exp((c + o(1))\sqrt{\log n})$ for almost all n , with $c = \sqrt{\log 2}$.

The first moment matches the lower bound: the expected number of balanced primes $p \leq e^t$ for random $n \leq N$ is governed by the endpoint of $\int (e^u/u) \exp(-a/u) du$, tending to 0 below $t = \sqrt{a}$ and to ∞ above — the same threshold constant $c = \sqrt{\log 2}$ from both sides. What remains for the second moment is pairwise decorrelation of balancedness at threshold depth:

$$\#\{n \leq N : n \in G_p \cap G_q\} \leq (1 + o(1)) \frac{|G_p \cap [1, N]| |G_q \cap [1, N]|}{N}, \quad p \neq q \text{ near } e^t.$$

This is a *finite-depth* two-base statement — far weaker than the uniform multi-base transversality that governs the companion problem #377 — and the empirical version holds with a bounded constant: the measured independence ratio for the pair (3, 5) oscillates in $[0.94, 1.82]$ over twelve scales with no trend, and for threshold-scale primes the digit depth is only $\Theta(\sqrt{\log n})$, opening the CRT window route (joint equidistribution of $(n \bmod p^i, n \bmod q^j)$ is exact while $p^i q^j \leq N$). One register note: a bounded constant in place of $(1 + o(1))$ yields, via Chebyshev, only a *positive proportion* of n with a near-threshold balanced prime; the almost-all conclusion needs the full asymptotic.

4 Numerical state

Instruments anchored on Kummer ($v_2 = s_2(n)$ exactly; 300 brute-force evaluations of $L(n)$ agree with the carry formula). Over $n \leq 4 \times 10^5$: the naïve exponent $\log L / \log n$ drifts downward (0.364, 0.311, 0.293, 0.285, 0.271 across dyadic ranges) while $\log L(n) / \sqrt{\log n}$ holds steady — the $\sqrt{\log}$ shape is clean. Deep scans (exhaustive to 10^8 ; 2×10^5 samples per decade to 10^{35} via 128-bit arithmetic) settle the constant: after removing the explicit second-order term $\frac{1}{2} \log \sqrt{a}$ of the law, the measured constant descends 0.935, 0.894, 0.859, 0.843 toward $\sqrt{\log 2} = 0.8326$, and the three-term form of the law predicts the raw mean at $\log n = 80.6$ to four decimals (0.9551 vs. 0.9550). A parameter-free independent-rail model (exact digit densities, product over rails) reproduces the median and the 5th/95th percentiles of the least balanced prime *prime-for-prime* in 14 of 15 decades — a joint test of rail independence far stronger than any marginal statistic. The minimum is achieved via the balanced channel in 81–89% of cases, by moderate primes ($p > 10$ in 74%, $p > 100$ in 1%), and the balanced channel dominates asymptotically (the $a \geq 2$ channels first fire at $\exp(2\sqrt{a})$ -scale).

5 Relation to #377

Both problems are governed by the balanced sets G_p . Problem #377 sums $1/p$ over balanced rails and needs uniform multi-base transversality; #731's answer is set by the *least* balanced

rail and needs only finite-depth pairwise decorrelation — the entry-level case of the same phenomenon. A proof of Conjecture 6 is therefore both a resolution of #731 and the first rigorous transversality brick for the harder problem.

References

- [1] P. Erdős, R. L. Graham, I. Z. Ruzsa, E. G. Straus, *On the prime factors of $\binom{2n}{n}$* , Math. Comp. 29 (1975), 83–92.
- [2] E. Kummer, *Über die Ergänzungssätze zu den allgemeinen Reciprocitätsgesetzen*, J. reine angew. Math. 44 (1852), 93–146.
- [3] C. Pomerance, *Remarks on the middle binomial coefficient*, note (two circulated versions, January 2026), math.dartmouth.edu/~carlp; journal reference to be confirmed.
- [4] T. F. Bloom, *Erdős problem #731*, erdosproblems.com/731 (accessed August 1, 2026).
- [5] R. C. Baker, G. Harman, J. Pintz, *The difference between consecutive primes, II*, Proc. London Math. Soc. (3) 83 (2001), 532–562.